

Non-Adiabatic Cocurrent Rotary Dryer Calculation Algorithm

Nomenclature:

CG: Air specific heat, $\text{J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

CS: Solids specific heat, $\text{J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

CW: Water specific heat, $\text{J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

D: Internal rotary dryer diameter, m

G: Air mass flow rate, dry basis, kg s^{-1}

GS: Air mass velocity, dry basis, $\text{kg s}^{-1} \text{ m}^{-2}$

H: Moist air enthalpy, J kg^{-1}

HS: Solids enthalpy, J kg^{-1}

L_{actual}: Known actual solids mass flow rate, dry basis, kg s^{-1}

L_{calc}: Calculated solids mass flow rate, dry basis, kg s^{-1}

LT: Length of a Transfer Unit, m

NT: Number of Heat Transfer Units

P: System total pressure, Pa

Pp: Water vapor partial pressure, Pa

Pss: Water vapor pressure at dry-bulb temperature, Pa

RH: Relative humidity, %/100

T: Moist air dry-bulb temperature, $^{\circ}\text{C}$

TS: Solids temperature, $^{\circ}\text{C}$ (TS1 - outlet; TS2 - inlet)

TW: Moist air wet-bulb temperature, $^{\circ}\text{C}$

Ua: Volumetric heat transfer coefficient, $\text{W m}^{-3} \text{ }^{\circ}\text{C}^{-1}$

W: Moist air absolute humidity, kg/kg

X: Solids moisture content, dry basis, kg/kg (X1 - outlet; X2 - inlet)

Z: Dryer shell length, m

α : Heat loss factor, dimensionless

K_{moist}: Empirical moisture model constant, $^{\circ}\text{C}^{-1}$

Equations:

1. **$\text{GS} = \text{G} / (\pi \times \text{D}^2 / 4)$**

2. **$\text{LT} = \text{GS} \times \text{CG} / \text{Ua}$**

3. **$\text{NT} = \text{Z} / \text{LT}$**

4. **$\text{W3} = 0.622 \times \text{Pp} / (\text{P} - \text{Pp})$, where $\text{Pp} = \text{RH} \times \text{Pss}$**

5. **$\text{G} \times (\text{W1} - \text{W2}) = \text{L}_{\text{actual}} \times (\text{X2} - \text{X1}_{\text{assumed}})$**

6. **$\text{HS} = (\text{CS} + \text{X}) \times \text{TS}$**

7. **$\text{G} \times (\text{H2} - \text{H4}) = \text{L}_{\text{actual}} \times (\text{HS4} - \text{HS2})$**

8. $G \times (H4 - H5) = L_{\text{actual}} \times (HS5 - HS4)$
9. $NT_{III} = ((T2 - T4) / ((T2 - TS2) - (T4 - TS4))) \times \ln((T2 - TS2) / (T4 - TS4))$
10. $NT_{II} = ((T4 - T5) / ((T5 - TS5) - (T4 - TS4))) \times \ln((T5 - TS5) / (T4 - TS4))$
11. $NT_I = NT - (NT_{III} + NT_{II})$
12. $\text{new}T1 = T5 - NT_I \times ((T5 - TS5) - (T1 - TS1)) / \ln((T5 - TS5) / (T1 - TS1))$
13. $L_{\text{calc}} = G \times (H2 - H1) / [(1 + \alpha) \times (HS1 - HS2)]$ (Non-adiabatic modification)
14. $X1_{\text{final}} = X2 \times \exp(K_{\text{moist}} \times (TS2 - TS1))$ (Final moisture after convergence)

Additional Relations:

- $W2 = W3$ (inlet air humidity equals ambient air humidity)
- $W4 = W2$ (humidity remains constant in section 2-4)
- $W5 = W1$ (outlet air humidity equals calculated $W1$)
- $TS4 \approx TW4$ (solids temperature at point 4 approximately equals wet-bulb temperature)
- $TS5 \approx TW5$ (solids temperature at point 5 approximately equals wet-bulb temperature)
- $T4 = (H4 - \lambda \times W4) / (CG + C_{pv} \times W4)$ (air temperature at point 4 from enthalpy)
- $T5 = (H5 - \lambda \times W5) / (CG + C_{pv} \times W5)$ (air temperature at point 5 from enthalpy)

Given Input Values:

$G = 5.444444444 \text{ kg/s}$
 $L_{\text{actual}} = 14.46388889 \text{ kg/s}$ (known actual flow rate)
 $W2 = W3 = 0.0106 \text{ kg/kg}$ (calculated from ambient conditions)
 $X2 = 0.066064913 \text{ kg/kg}$
 $X1_{\text{assumed}} = 0.001 \text{ kg/kg}$ (fixed assumption during iteration)
 $T2 = 980^\circ\text{C}$
 $TS2 = 26^\circ\text{C}$
 $TS1 = 186.09^\circ\text{C}$ (current guess - outer loop variable)
 $CS = 880 \text{ J/(kg}\cdot^\circ\text{C)}$
 $H2 = 1,114,179 \text{ J/kg}$ (calculated from $T2$ and $W2$)
 $NT = 3.1710$
 $CG = 1109.5 \text{ J/(kg}\cdot^\circ\text{C)}$
 $\alpha = 0.2$ (20% heat loss factor)
 $K_{\text{moist}} = 0.0338 \text{ }^\circ\text{C}^{-1}$ (moisture model constant)

Current Iteration Values (Example):

$TW4 = 50^\circ\text{C}$ (inner loop variable)
 $TW5 = 45^\circ\text{C}$ (inner loop variable)

T1 = 60°C (inner loop variable)

W1 = 0.2137 kg/kg (calculated from water balance)

Algorithm Steps:

Outer Loop: Iterate for Outlet Solids Temperature (TS1)

Step 1: Calculate W1 using equation 5

$$W1 = W2 + [L_{\text{actual}} \times (X2 - X1_{\text{assumed}})] / G$$

Step 2: Calculate solids enthalpies using equation 6

$$HS2 = (CS + X2) \times TS2$$

$$HS1 = (CS + X1_{\text{assumed}}) \times TS1 \text{ (current guess)}$$

Inner Loop 1: Iterate for Wet-Bulb Temperature at Point 4 (TW4)

Step 3: Calculate HS4 using current TW4

$$TS4 = TW4 \text{ (model hypothesis)}$$

$$HS4 = (CS + X2) \times TS4$$

Step 4: Calculate H4 using equation 7

$$H4 = H2 + [L_{\text{actual}} \times (HS4 - HS2)] / G$$

Step 5: Calculate air temperature T4

$$T4 = (H4 - \lambda \times W4) / (CG + C_{pv} \times W4) \text{ where } W4 = W2$$

Step 6: Request psychrometric data for new TW4

Based on: Total enthalpy H4, Air temperature T4, Humidity W4

Calculate TW4_new from psychrometric properties

Step 7: Check TW4 convergence

If $|TW4 - TW4_{\text{new}}| < \epsilon$, proceed to Step 8. Else, update TW4 and return to Step 3.

Inner Loop 2: Iterate for Wet-Bulb Temperature at Point 5 (TW5)

Step 8: Calculate HS5 using current TW5

$$TS5 = TW5 \text{ (model hypothesis)}$$

$$HS5 = (CS + X1_{\text{assumed}}) \times TS5$$

Step 9: Calculate H5 using equation 8

$$H5 = H4 + [L_{\text{actual}} \times (HS5 - HS4)] / G$$

Step 10: Calculate air temperature T5

$$T5 = (H5 - \lambda \times W5) / (CG + C_{pv} \times W5) \text{ where } W5 = W1$$

Step 11: Request psychrometric data for new TW5

Based on: Total enthalpy H5, Air temperature T5, Humidity W5

Calculate TW5_new from psychrometric properties

Step 12: Check TW5 convergence

If $|TW5 - TW5_new| < \epsilon$, proceed to Step 13. Else, update TW5 and return to Step 8.

Inner Loop 3: Iterate for Outlet Air Temperature (T1)

Step 13: Calculate NTU for each zone using equations 9, 10, and 11

$$NT_{III} = ((T2 - T4) / ((T2 - TS2) - (T4 - TS4))) \times \ln((T2 - TS2) / (T4 - TS4))$$

$$NT_{II} = ((T4 - T5) / ((T5 - TS5) - (T4 - TS4))) \times \ln((T5 - TS5) / (T4 - TS4))$$

$$NT_I = NT - (NT_{III} + NT_{II})$$

Step 14: Calculate new T1 using equation 12

$$T1_new = T5 - NT_I \times ((T5 - TS5) - (T1 - TS1)) / \ln((T5 - TS5) / (T1 - TS1))$$

Step 15: Check T1 convergence

If $|T1 - T1_new| < \epsilon$, proceed to Step 16. Else, update T1 and return to Step 14.

Final Calculations and Outer Loop Check

Step 16: Calculate H1 using psychrometric properties

Request dry air enthalpy at T1, then:

$$H1 = H_{dry,1} + W1 \times (\lambda + C_{pv} \times T1)$$

Step 17: Calculate L_calc using equation 13 (Non-adiabatic)

$$L_calc = G \times (H2 - H1) / [(1 + \alpha) \times (HS1 - HS2)]$$

Step 18: Check TS1 convergence (Outer Loop)

$$L_error = |L_actual - L_calc| / L_actual \times 100\%$$

If $L_error < \epsilon_L$, proceed to Step 19. Else, adjust TS1 and return to Step 1.

Step 19: Calculate final moisture content using equation 14

$$X1_final = X2 \times \exp(K_moist \times (TS2 - TS1_converged))$$

Step 20: Algorithm complete - Results:

TS1 = converged outlet solids temperature

X1 = X1_final (use this, not X1_assumed)

T1 = converged outlet air temperature

Important Notes on Psychrometric Data:

During the iteration process, the algorithm requires psychrometric data at specific points:

Point 4 (TW4 Calculation):

- **Input:** Total enthalpy H4, Air temperature T4, Humidity W4
- **Request:** New wet-bulb temperature TW4

Point 5 (TW5 Calculation):

- **Input:** Total enthalpy H5, Air temperature T5, Humidity W5
- **Request:** New wet-bulb temperature TW5

Point 1 (H1 Calculation):

- **Input:** Air temperature T1
- **Request:** Dry air enthalpy at T1
- **Calculate:** $H1 = H_{dry,1} + W1 \times (\lambda + C_{pv} \times T1)$

Key Modifications for Non-Adiabatic Operation:

1. **Heat Loss Factor (α):** Accounts for 20% heat loss to environment
2. **Modified Energy Balance:** L_{calc} includes $(1 + \alpha)$ factor in denominator
3. **Moisture Model Timing:** $X1_{assumed}$ used during iteration, $X1_{final}$ calculated after TS1 converges
4. **Dual Temperature Tracking:** Both air temperatures (T4, T5) and wet-bulb temperatures (TW4, TW5)

Convergence Criteria:

- **TW4, TW5 errors:** $< 0.01^{\circ}\text{C}$
- **T1 error:** $< 0.01^{\circ}\text{C}$
- **L_error:** $< 0.1\%$
- **All must be satisfied simultaneously**

Current Status:

Algorithm is at the beginning of outer loop iteration

Proceed with calculating W1 using current TS1 = 186.09°C

Expected Results (After Convergence):

- **Outlet solids temperature (TS1):** Converged value
- **Outlet moisture content (X1):** $X1_{final}$ from empirical model
- **Outlet air temperature (T1):** Converged value
- **Heat duty to solids:** $L_{actual} \times (HS1 - HS2)$
- **Heat lost to environment:** $\alpha \times Q_{solids}$
- **Total heat required:** $(1 + \alpha) \times Q_{solids}$